Report

Of
Financial Cybersecurity Web Platform & Secure Cloud Architecture

Submitted

To

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B. Tech CSE

By

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CANDIDATE'S DECLARATION

I, Shubham Chaudhary do hereby declare that the internship report titled "Financial Cybersecurity Web Platform & Secure Cloud Architecture" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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2. Executive Summary

This report documents the engineering tasks, methodologies, and technical deliverables completed during the Commonwealth Bank Software Engineering Virtual Internship program. The primary objective was to architect, style, and outline a secure web-based digital client education platform focused on financial cybersecurity. The project simulated real-world enterprise software engineering workflows at a premier banking institution.

Throughout the internship, four core milestones were executed: foundational markup structuring using HTML5, client-centric cybersecurity content synthesis, enterprise brand alignment using CSS3 according to strict Commonwealth Bank styling standards, and the formulation of a comprehensive cloud hosting and infrastructure security proposal compliant with international banking regulations (APRA CPS 234).

Key Project Outcomes:

- Engineered a fully accessible, semantic HTML5 client security portal.
- Authored actionable, non-technical cybersecurity guidance across five critical vectors (Authentication, MFA, Phishing Defense, Network Encryption, and Device Hygiene).
- Implemented responsive, strict brand-compliant typography and visual hierarchy using corporate palette rules.
- Formulated a hardened enterprise hosting architecture incorporating TLS 1.3, Layer 7 WAF, DDoS mitigation, and Zero-Trust access protocols.

3. Internship Context & Organization Profile

Commonwealth Bank of Australia (CommBank) is one of the leading integrated financial services organizations in the Asia-Pacific region. As digital banking adoption continues to accelerate, the intersection of user experience, software engineering, and proactive cybersecurity becomes pivotal.

The Software Engineering Program exposes candidates to real-world engineering problem statements, simulating the duties of an entry-level software engineer. Engineers must balance technical robustness with accessible communication and enterprise design consistency to serve millions of retail and commercial banking clients.

4. Milestone Breakdown & Technical Execution

The project was divided into four sequential sprints, each representing key phases in frontend development, content architecture, brand integration, and cloud operations.

Milestone / Task	Key Objective	Deliverable & Technology
Task 1: Basic Web Page Setup	Initialize local development environment, establish markup architecture, and link external stylesheets.	index.html, index.css (HTML5/CSS3 boilerplate setup).
Task 2: Content Synthesis	Translate complex cybersecurity principles into clear, digestible guidelines for banking customers.	Structured semantic copy (~350 words) organized with <h2> and <p> tags.
Task 3: Brand Styling Compliance	Incorporate CommBank corporate brand guidelines, color palettes, and typography constraints.	Custom CSS styling utilizing Primary Yellow (#FFCC00), Beacon Yellow (#FFFF00), and Neutral scales.
Task 4: Web Hosting Proposal	Devise a multi-tier cloud hosting architecture emphasizing data integrity, encryption, and DDoS defense.	Comprehensive technical proposal addressing TLS 1.3, WAF, CI/CD security, and APRA compliance.

Detailed Sprint Execution Summary

- Sprint 1 (Environment & Markup Setup): Established the foundational workspace in Visual Studio Code. Created modular file structures with normalized box-sizing models (* { box-sizing: border-box; }) to ensure consistent box model rendering across different browser rendering engines.
- Sprint 2 (Content Integration): Conducted domain research into contemporary threat vectors facing retail banking clients, specifically targeted credential stuffing, SIM swapping, phishing, and man-in-the-middle exploits over unsecured Wi-Fi.

5. Security Content & Client Education Strategy

In digital financial applications, security engineering is only as strong as user awareness. The synthesized cybersecurity curriculum embedded into the portal addresses five core attack surfaces:

- 1. Credential Hardening: Recommends multi-word passphrases combining unrelated words, numbers, and symbols to resist brute-force and dictionary attacks, emphasizing zero cross-account reuse.
- 2. Multi-Factor Authentication (MFA): Mandates hardware security keys or timebased one-time password (TOTP) authenticator applications over legacy SMS verification to eliminate SIM swap vulnerabilities.
- 3. Phishing & Social Engineering Defense: Educates users on communication verification protocols, highlighting that authentic banking systems never solicit PINs, passwords, or full credentials via unsolicited SMS/email.
- 4. Transport Layer Awareness (Public Wi-Fi): Highlights interception risks on unencrypted networks and guides users toward cellular connections or enterprisegrade VPN tunnels.
- 5. Endpoint Hygiene & Monitoring: Promotes automated OS/browser patching and real-time push alert configuration for immediate unauthorized debit detection.

6. Branding Implementation & Code Architecture

To maintain enterprise brand coherence, the front-end layout strictly adheres to the Commonwealth Bank Style Guidelines:

- Typography Standards: Unified sans-serif family ('Gill Sans', 'Lucida Sans', sansserif) with strict proportional scaling: Title at 70pt, Subheadings at 38pt, and Body at 18pt.
- Color Palette Rules: Primary Yellow (#FFCC00) for primary visual focus, Beacon Yellow (#FFFF00) for warning highlights, Light Gray (#EBEBEB) for background canvas, and Black (#1E1E1E) with pure White (#FFFFFF) text for high-contrast dark sections.

```
/* Commonwealth Bank Brand-Compliant CSS Excerpt */ body {   background-color:
#EBEBEB;   color: #1E1E1E;   font-family: "Gill
Sans", "Lucida Sans", sans-serif;   font-size: 18pt;
}   h1 { font-size: 70pt; color: #1E1E1E; } h2 { font-
size:
38pt; color: #1E1E1E; }
.card-dark {
  background-color: #1E1E1E;
  color: #FFFFFF; /* High-contrast white text on dark background */   border-left: 12px solid #FFCC00;
}
```

7. Enterprise Web Hosting & Security Proposal

While the initial portal serves static informational assets, enterprise financial systems must be designed for secure scalability. The hosting proposal outlines a defense-in-depth infrastructure topology:

- Edge Protection & Content Delivery: Global CDN (AWS CloudFront / Cloudflare) coupled with a Web Application Firewall (WAF) to inspect Layer 7 HTTP/HTTPS traffic and absorb Layer 3/4 volumetric DDoS attacks.
- Encrypted Transport: Strict TLS 1.3 enforcement with automated certificate renewal via Let's Encrypt / AWS ACM and HTTP Strict Transport Security (HSTS) preloading to preclude SSL stripping.

- Immutable Storage & Serverless Delivery: Static assets hosted in an isolated AWS S3 bucket with public access blocked, delivered strictly through CloudFront Origin Access Control (OAC).
- CI/CD & DevSecOps Automation: Automated pipeline deployment via GitHub Actions integrating static application security testing (SAST) and automated dependency vulnerability checks (Dependabot/Snyk).

8. Regulatory Compliance & Risk Assessment

Threat / Challenge	Potential Impact	Architectural Mitigation
Data Interception (MitM)	Client credential theft on public networks.	Enforce TLS 1.3 with 256-bit AES encryption & HSTS preloading.
Volumetric DDoS Attacks	Service unavailability and brand reputational damage.	Anycast CDN rate-limiting, geo-blocking, and Layer 4 automated scrubbing.
Privilege Escalation	Unauthorized modi ication of web portal content.	Granular IAM policies, mandatory hardware MFA, and ephemeral CI/CD tokens.
Regulatory Non-Compliance	Fines under APRA CPS 234 & Privacy Act standards.	Immutable audit logging (CloudTrail), automated compliance posture monitoring.

9. Learning Outcomes & Conclusion

The Commonwealth Bank Software Engineering Virtual Internship provided rigorous exposure to enterprisestandard engineering practices. Beyond syntax and styling, the experience underscored the critical responsibility software engineers hold in safeguarding digital trust within financial ecosystems.

Summary of Key Competencies Acquired:

- Front-End Engineering: Mastered semantic HTML5 document structuring and CSS3 modular styling adhering to strict corporate design tokens.
- Technical Communication: Cultivated the ability to translate technical security protocols into accessible, empowering guidance for non-technical retail banking clients.
- Cloud Infrastructure Design: Developed architectural competency in architecting highly available, hardened cloud hosting environments with zero-trust principles.
- Regulatory Governance: Gained awareness of banking sector compliance frameworks (APRA CPS 234) and enterprise security baselines.